

## Distributive & Expanding — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

### Objective

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**Q1** (b)  $ab + ac$ . Distributive property.

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**Q2** (b)  $ab + an + mb + mn$ . Four products.

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**Q3** 15.  $5 \times 3 = 15$ .

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**Q4** (a)  $15p^2 - 6pq$ .  $(-3p)(-5p) = 15p^2$ ;  $(-3p)(2q) = -6pq$ .

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**Q5** False. Different powers of the letters, so not like terms.

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**Q6** (a) Both true, R explains A.  $23 \times 27 + 23 = 23(27 + 1) = 23 \times 28$ .

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### Short answer

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**Q7** (i)  $uv - 3u + 3v - 9$  (ii)  $yz + 5y + 9z + 45$ .

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**Q8**  $100ac + 10ad + 10bc + bd$ .

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**Q9**  $5a^2b + 4ab^2$ .  $3a^2b + 2a^2b$  combine;  $4ab^2$  stays separate.

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### Long answer

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**Q10**  $-x^2 + 9x - 18$ .  $3x - 18 - x^2 + 6x$ , then combine  $3x + 6x$ .

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**Q11**  $4a + 5ab + ab^2 - 12b^2 - 3b^3$ . Multiply each term of the first bracket by  $(4 + b)$  and combine  $ab + 4ab$ .

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**Q12**  $4y^2 + 44yz - 5y + 77z - 21$ . Combine  $-12y + 7y = -5y$ .

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Total: 21 marks.